

Zakariae El Mernissi

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Professional Experience

GenAI Consultant, Deloitte

12/2025 – Present
Casablanca

- **Client Engagement: Technip Energies**
- **Autonomous Incident Resolution & Remediation:** Engineered a multi-agent system that autonomously investigate the exact root cause and location of production errors, automate remediation, and apply fixes through a mandatory human validation gate for security and compliance. This significantly reduces Mean Time to Repair (MTTR), ensures operational continuity of the run team.

Specialized Agent Architecture: Architected dedicated agent roles a KB Agent for instant solution matching, an OCR Agent for screenshot analysis, a Log Agent for deep technical troubleshooting, and a Web Search Agent for real-time documentation retrieval enabling autonomous diagnosis and controlled remediation.

Self-Improving Knowledge Flywheel: Implemented a sophisticated feedback loop that autonomously enriches the Knowledge Base (KB) with human-validated resolutions, creating a continuously self-improving system that grows in accuracy and domain expertise over time.

Tech: Python, Azure AI Search, Azure OpenAI, Claude, Azure AI Foundry, Agent skills, Microsoft Agent framework, ServiceNow ticket, Logic app

GenAI Engineer, Capgemini TS

03/2024 – 12/2025
Casablanca, Morocco

- Member of the GenAI Lab

- **AI-Powered HR Platform:** Intelligent CV Management System

Designed and deployed a production multi-agent GenAI copilot using Google's Agent Development Kit (ADK), enabling automated resume parsing and intelligent candidate-role matching, reducing manual screening effort by an estimated ~75–80% .

Translated HR operational requirements into scalable AI copilot specifications, defining agent roles, data flows, and processing pipelines for end-to-end recruitment automation

Engineered optimized conversation flows and prompt engineering strategies that improved response relevance and reduced interaction friction across recruiter-facing interfaces

Integrated Vertex AI with Cloud Storage and PostgreSQL to support structured candidate data retrieval and persistent session management

- **Tech:** Python, GCP ADK, Gemini, Vertex AI, Cloud Storage, PostgreSQL, MarkitDown

- **Supply Chain Analyzer:** GenAI-Powered Logistics Intelligence System:

Built a cloud-native conversational AI system on AWS Bedrock for real-time analysis of international supply chain constraints, enabling ~60–65% **faster** query resolution compared to manual document review

Architected a full RAG pipeline using Amazon Titan for document embeddings and OpenSearch Service as the vector store, supporting multi-source data ingestion and semantic search at scale

Implemented Claude via Amazon Bedrock as the reasoning layer, delivering contextual AI responses with cited document references improving analyst trust and auditability

Designed API configurations and data flow integration specifications across Lambda, API Gateway, S3, CloudFront, and Cognito for a secure, scalable serverless architecture

- **Tech:** Python, AWS Bedrock, Claude, Amazon Titan, OpenSearch, AWS Lambda, Amazon API Gateway, Amazon S3, CloudFront, Cognito

- **Computer System Validation (CSV) Platform:** Pharmaceutical Industry :

Engineered a multi-agent GenAI system using CrewAI with specialized agent roles for document analysis and compliance case management, reducing pharmaceutical validation cycle time by an estimated ~45–55%

Translated pharmaceutical regulatory requirements into structured technical specifications, defining agent responsibilities, validation logic, and reporting workflows aligned with GxP compliance standards

Implemented advanced prompt engineering strategies, including zero-shot and few-shot learning to ensure accurate, regulation-aware outputs across validation document types

Established CI/CD deployment pipelines via GitHub Actions, enabling automated testing and continuous delivery to Azure Web Apps with minimal downtime

- **Tech :** Python, Azure AI Services, Openai, CrewAI, AI Search, MarkitDown, GitHub Actions, Azure Web Apps

- **Codebase Analysis & Documentation Generator:** Intelligent Code Understanding Copilot:

Built a dual-path code intelligence system combining Pinecone vector embeddings and Neo4j knowledge graph construction, enabling semantic code search and relationship mapping across complex codebases reducing documentation and onboarding time by an estimated ~55–65%

Automated directory and file extraction with entity identification for code components, dependencies, and architectural relationships, producing structured knowledge graphs for downstream querying

Implemented a Text-to-Cypher natural language interface, allowing non-technical stakeholders to query codebase structure and retrieve technical insights without writing code

Applied recursive chunking and LangChain orchestration to optimize embedding quality and maintain contextual coherence across large repository processing

- **Tech** : Python, Pinecone, Neo4j, Gemini, google embeddings, chunking, Langchain.

Data scientist · Stage PFA, Office National des Chemins de Fer (ONCF)

08/2023 – 09/2023

- Data Analysis & System Migration:

Rabat, Morocco

Analysis of supplier and item data from the Procurement Information System.

ETL pipelines for migration preparation.

Analytical dashboards with Python (**Pandas, NumPy**) for procurement insights visualization.

- **Tech stack**: Python, Pandas, NumPy, Microsoft SQL Server, Power bi.

Freelance

Machine Learning Engineer

Morocco, Remote

- Personalized Learning Analytics System

Built a machine learning system that analyzes student behavior on an e-learning platform.

Used student onboarding forms (ILS and HEXAD) to determine learner profiles and motivations.

Designed a tracking system for course progress, quiz attempts, and interactions with motivational messages.

Implemented feedback loops using "Like/Dislike" reactions to messages, enabling adaptive encouragement strategies.

Provided actionable insights on student engagement and personalized learning experiences.

- **Tech stack**: Python, Pandas, Scikit-learn, Streamlit, SQLite, JSON, Matplotlib

GenAI Solutions Architect & Automation Engineer

Canada, Remote

- Creative Intelligence Ecosystem & Enterprise Automation (Le Grand Chien)

Architected a centralized "GPT HUB" orchestrating specialized AI agents designed to support the four pillars of the agency's creative lifecycle: Exploration, Co-creation, Creation, and Radiation.

Developed "Backstage" intelligent assistants to automate administrative workflows, including financial tracking (invoicing, grants), internal project coordination, and strategic business development analysis.

Engineered complex automation pipelines connecting Custom GPTs with enterprise ecosystems (Notion, Zoho, Google Suite, Meta) using Make and REST APIs for seamless data synchronization.

Designed intelligent workflow structures ensuring tone consistency and sensitive intelligence across all agents.

- **Tech stack**: Python, OpenAI API, Perplexity, Claude, Make, Notion API, Zoho CRM, REST APIs, Prompt Engineering.

GenAI/Prompt Engineer

France, Remote

- AI Agent System for HR Recruitment (SplendUp)

Developed intelligent multi-agent system using CrewAI framework with Claude Sonnet 4 for automated candidate identification and outreach automation

Integrated Coresignal MCP (Model Context Protocol) for real-time decision-maker discovery and professional network intelligence gathering

Implemented specialized agent roles for candidate research, company analysis, and personalized message generation

Designed advanced prompt engineering strategies for context-aware outreach message creation tailored to specific job openings and candidate profiles

Built automated workflow for recruitment consultants to streamline candidate promotion process from identification to personalized communication

Created intelligent decision-maker targeting system analyzing company hierarchies and organizational structures

Developed personalized messaging engine generating role-specific outreach content based on candidate background and position requirements

Tech stack: Python, CrewAI, Claude Sonnet 4, Coresignal MCP, Prompt Engineering, Multi-Agent Architecture

GenAI Engineer

USA, Remote

- AI Agent for Commercial Property Ownership Research(Upwork)

Developed an intelligent AI agent system for automated commercial property ownership research and due diligence

Built multi-source data integration pipeline connecting public records, property databases, and legal document repositories

Implemented entity extraction and relationship mapping using GPT-4 for property ownership chains and corporate structures

Created automated web scraping workflows with intelligent data validation and cross-referencing capabilities

Designed knowledge graph representation using Neo4j to visualize complex ownership relationships and property portfolios
Built semantic search capabilities for property records using vector embeddings and contextual retrieval
Developed comprehensive reporting system with automated property analysis, risk assessment, and ownership timeline generation
Integrated compliance checks for regulatory requirements and due diligence standards
- **Tech stack**: Python, OpenAI GPT-4, Neo4j, BeautifulSoup, LangChain, Faiss, FastAPI, PostgreSQL

GenAI Engineer

USA, Remote

- English Accent Analyzer : AI-Powered Multilingual Audio Intelligence System (Upwork)

Built comprehensive video-to-audio processing pipeline supporting YouTube URLs and direct video file formats.
Implemented advanced audio preprocessing with noise reduction using Deep Filternet for enhanced accent detection accuracy.
Integrated Wav2Vec2 transformer model via Hugging Face for sophisticated accent classification (British, American, Australian, etc.)
Created confidence scoring system with AI-powered natural language explanations using OpenRouter API integration.
Designed user-friendly interface with real-time processing capabilities and detailed analytics dashboard.
Automated multimedia content analysis with seamless video download and audio extraction workflows
- **Tech stack**: Python, Streamlit, Hugging Face Transformers (Wav2Vec2), MoviePy, LibROSA, Deep Filternet, yt-dlp, OpenRouter API (Mistral), Audio Processing Libraries

Technical Skills

Generative AI

- **LLMs**: OpenAI GPT models, Google Gemini , Claude, Mistral & others
- **Model Integration**: LiteLLM, Portkey, OpenRouter, Langgraph
- **LLM Frameworks**: LangChain, Langgraph, LlamaIndex
- **Multi-Agent AI** : CrewAI, Google ADK, Autogen, Semantic Kernel,
- **Vector Databases**: Vertex AI Vector Search, Pinecone, AWS OpenSearch Service, Weaviate, Qdrant, ElasticSearch , Azure AI Search
- **Chunking**: Chonkie(Framework), Recursive chunking, Semantic chunking, context-aware
- **Parsing** : Docling, Markitdown, Unstructured, pdfplumber, PyMuPDF

Cloud Platforms & AI Services

- **Microsoft Azure**: AI Services, Azure OpenAI, Azure Blob Storage, Azure AI Foundry, Azure AI Search
- **Google Cloud Platform**: Vertex AI, Cloud Run, Cloud Storage, BigQuery, GCP ADK .
- **Amazon Web Services**: Bedrock, Lambda, API Gateway, S3,

Python Development & APIs

- **Programming**: Python, SQL
- **API Development**: FastAPI, Flask
- **UI Frameworks**: Streamlit, Gradio
- **Package Management**: Poetry , virtual environments

Database Management

- MongoDB, Neo4j, PostgreSQL

DevOps & Infrastructure

- **CI/CD**: GitLab CI/CD , Github actions
- **Version Control**: Git
- **Infrastructure as Code**: Terraform

AI Evaluation & Testing

- **Evaluation Frameworks**: DeepEval, RAGAS, TruLens, MLFlow
- **Metrics**: Faithfulness, answer relevancy, retrieval accuracy, hallucination detection

Certifications

- OCI Generative AI Professional – Oracle
- Microsoft Azure AI Fundamentals
- AWS Partner: Generative AI Essentials
- Google Cloud : Generative AI

Hackathon

Google cloud GenAI Hackathon 2024

Microsoft Generative AI Hackathon 2025

AWS Generative AI Hackathon 2024

Education

Master's in Mobility and Big Data , <i>Faculty of Sciences and Technologies</i>	09/2022 – 07/2024 Tangier, Morocco
Bachelor's in computer engineering , <i>Faculty of Sciences and Technologies</i>	10/2021 – 07/2022 Tangier, Morocco
Diploma of Scientific and Technical University Studies (DEUST): MIP , <i>Faculty of Sciences and Technologies</i>	09/2018 – 06/2021 Tangier, Morocco

Languages

English: Advanced — EFSET English Certificate (CEFR Level C1) • **French: Advanced**